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An idea to explore: Using origami to learn molecular structure of biomolecules

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Abstract

The COVID-19 pandemic affected a large range of in-person education activities in Colombia. This created great limitations in academic performance for students with reduced access to communication technologies and deepened the educational gaps in the country. This was particularly true for sciences such as biochemistry. In Colombia, molecular structure is a subject traditionally taught through 2D drawings and static diagrams because software and 3D artifacts are not available to all students. Thus, it is essential to develop and apply strategies to study molecular structure; especially tools that are accessible and can be easily built and used at home in rural areas of the country. Here, we propose the use of origami as a tool to teach molecular structure to second year college students in Colombia. We describe the development and the implementation of the tool adjusted to students' needs regarding their visual, tactile, and other experiential learning. We included serious game elements during the implementation to engage participation and teamwork. Students' perception about the use and utility of origami to study molecular structure was favorable, highlighting its simplicity and powerfulness to help them grasp key concepts in chemistry. This motivates us to propose this idea to explore and continue improving the strategy in the classroom.

KEYWORDS

paper, serious game, structure and function, visual literacy

1 | INTRODUCTION

In Colombia, the COVID-19 pandemic affected all activities related to in-person education. The governments and the directors of the schools opted to continue the training processes through remote schemes, which led to the revision of the curricula and the design of teaching strategies to guarantee the education of children and young people.[1,2] However, the lack of access to information and communication technologies for a significant proportion of students had consequences on their training processes and on their academic performance, which deepened the educational gaps.[2] During COVID-19 pandemics biochemistry courses were taught online forcing instructors to look for new methods to study molecular structure adjusted to the resources available to the local context, thus we implemented origami and identified it as a valuable idea to explore not only online, but also in-person teaching. We present the preliminary results of the implementation of this idea with students majoring in Biology at the Universidad Nacional de Colombia Sede de La Paz.

1.1 | Origami as a tool to understand molecular structures

Origami, the art of folding a two-dimensional (2D) paper into a three-dimensional (3D) model, originated in Japan

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during the Edo period, possibly as a ceremonial craft.[3] The folding principles of Origami reflect those of the structural components observed in various fields of study including biochemistry.[3,4] For instance, the 3D structure of macromolecules results from folding a single chain of monomers, likewise, many 3D models of origami can be formed from a single sheet of paper.[3]

Understanding the molecular structure of biomolecules is critical to the study of biochemistry, because biological function depends on molecular structure.[5] Reality is 3D, and thus, life is based in 3D molecules. However, in the classrooms, biochemistry is taught and learned in two dimensions. Texts and the blackboard are used to explain biomolecules in 2D without going beyond static descriptions and diagrams. Several advancements in educational technology have allowed the incorporation of virtual didactic tools where it is possible to visualize structures in 3D, rotate molecules and observe them from different angles, but these tools are still 2D (the screen of a computer or mobile device).[6] To overcome this limitation, several types of 3D models have been developed, mostly made of plastic or Styrofoam. These didactic models are expensive and non-biodegradable.[7] The use of paper-folded origami provides a cheaper, accessible, and versatile alternative to visualize and manipulate the 3D structures of biomolecules. Students traditionally study molecular structure in 2D, transitioning to a 3D model is a higher level of representation that uses additional senses (touch). Also, students have unlimited access to pieces and develop new structures depending on their interest. Furthermore, this strategy may include gamification elements that serve as a hook to capture the interest of students by being interactive, fun, and challenging beyond the school environment.[8]

Origami also allows developing spatial perception skills, that is, taking structures from a plane and being able to visualize them in three dimensions. This ability is part of what has been termed by many authors as visual literacy.[9] It is defined as the ability to understand (read) and to use (write) pictures, as well as to think and to learn in terms of images. Visual literacy has been considered a core competency in life sciences education.[10] To engage students in activities that include visual literacy in biochemistry, teachers use different types of visualization tools (external representations) such as physical models, photographs, micrographs, drawings, diagrams, images, analog representations, metabolic maps, symbolic pathways, genomic representations, graphs, icons, multimedia, and augmented and virtual reality environments. All of these are aimed at establishing spatial relationships to build mental models to understand biochemical phenomena.[11] The visualization process has three main stages: internalization (process by which the sense organs work with the brain to absorb information from the world), conceptualization (process by which meaning is created and cognitive models are built) and externalization (production of external visual models).[12]

Origami can be used to learn concepts such as chirality, the three-dimensional nature of the biochemical environment in the cells, and enzyme-substrate or ligand-receptor molecular interactions. Teaching biochemistry using origami allows the students to understand why the functional architecture of a biomolecule requires a precise folding sequence within a cellular context.[13] This correct folding sequence is vital to maintaining structure and biological activity; if errors occur, they can cause dire consequences as in the case of prions.[8,14] In addition, molecular structures are not static, they interact with other biomolecules to exchange matter, energy and information through metabolism. Origami makes it possible to address the challenges of teaching metabolism, since it favors kinesthetic learning (tactile experience) of biomolecular interactions.[4,15]

1.2 | Implementing origami in the classroom to visualize molecular structure

As a strategy to teach molecular structure and strengthen the competences in visual literacy for students, here, we propose the use of origami, assessing students' perceptions about its use and utility in the classroom.

In this paper, we describe one example used in the course "Principles of Biochemistry" for Biology students at the Universidad Nacional de Colombia, Sede de La Paz. The learning objectives and competences for the course are (1) to describe the molecular structure of biomolecules (2) to identify the function of biomolecules and (3) to understand the flow of energy and matter in biological systems through metabolic pathways. Origami is used as a tool to achieve objective 1, it is implemented in three different sessions of 2 h during 3 weeks. The goal of Session 1 is to describe and understand the atom's structure. The goal of session 2 is to describe the basic structure of simple molecules and session 3 has the goal to understand the structure of complex molecules (Table 1).

To achieve these goals, gamification elements are included in the form of a serious game intended to develop students' knowledge, skills, and attitude competences. Game is based on a competition to obtain score points. Session 1 is individual, and session 2 and 3 are developed in teams. To obtain one point a student/team should be the fastest to correctly finish the activity proposed by the instructor. At the end of each session, it is expected that each student/team has built each of the

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TABLE 1 Session topic, goals, and gamification elements for the proposed strategy.

Session topic	Goal	Preliminary concepts	Activities and game scoring system (points available)	Supportive digital tools
SESSION 1: Bioelements	Building the six primary bioelements (C,H,O, N,P, S) from building blocks	Matter structure: Atom, element	Individual activity. Fold paper into a Prism (1) Fold paper into a Connector (1) Use prisms and connectors to build Bioelements (6)	Google's periodic table Google interactive: https://artsexperiments.withgoogle.com/periodic-table/
SESSION 2: Simple molecules	Building simple biomolecules (water, methane, amoniac, carbon dioxide, sulfhydric acid, phosphine) from building blocks	Chemical bond, valence electrons, molecule	Group activity: Fold paper to build a link (1) Use bioelements built in previous session to build simple molecules (6)	Organic chemistry made transparent. Universidad Autónoma de Madrid http://qorganica.es/QOT/jsmol/panel_jme.htm
SESSION 3: Complex molecules	Building monomers of complex biomolecules (ribose, propanoic acid, glycine, cysteine, adenine, adenosine monophosphate nucleotide)	Functional groups, monomers and polymers	Group activity: Use simple biomolecules built in previous session to build complex molecules (2)	Chemspider https://www.chemspider.com/StructureSearch.aspx

building blocks, elements or molecules proposed and has taken a photograph that is submitted to the instructor. This can be done in class or after class in case students were not able to finish during class time. The activities and game scoring system are presented in column 4 of Table 1.

1.2.1 | First session: Building bioelements from construction blocks

For the construction of origami bioelements, the Corey, Pauling and Koltun (CPK) color scheme is used. In this scheme the atoms are represented by the following colors: Carbon (gray), hydrogen (white), oxygen (red), nitrogen (blue), sulfur (yellow) and phosphorus (orange). The folding instructions are taken from Garrido.[5] However, in her proposal, she describes the chemical elements as a tetrahedral (carbon structure), her model does not consider that each element has a different value of valence electrons. This is not trivial, since this parameter defines the spatial distribution of atoms and how they relate to each other to make biomolecules, shaping their tridimensional structure. Thus, we propose facilitating the visualization of valence electrons by using an alternative organization of Garrido's three basic types of building blocks: prismatic pieces, connectors, and links. This is a critical aspect of our proposal, since we aim to represent the 3D structure of biomolecules using the most faithful model to its actual structure using Origami.

The first session involves building the primary bioelements (C, H, O, N, P, S) using building blocks. Students observe an IUPAC-color coded 3D molecular representation of a popular biomolecule (such as glucose or cholesterol) and are introduced to the color nomenclature for bioelements (i.e. gray = carbon, red = oxygen, white = hydrogen). Prior to this session students were introduced to the topic "Structure of matter: atoms, elements and molecules" and were instructed to bring paper squares (5 × 5 cm) in white, red, gray, blue, orange, and yellow to the classroom.

The first game element consists of a competition in which individual students are prompted to build a prism with a piece of paper of their choice following the instructions shown on a screen using the image in Figure 1a. The winner is the fastest student to build a correct prism. The winner demonstrates step by step to the class how he/she built the prism until everyone in the class has the piece built. Immediately after all students have built their prism, the teacher explains that each prism represents an available electron to create a chemical bond. The color of the paper represents which bioelement it belongs to. Immediately after, the game is repeated following the image in Figure 1b that represents a connector. The instructor explains that this represents the electromagnetic force that keeps the electrons around the atom nucleus. Next, students are exposed to the periodic table (google interactive: https://artsexperiments.withgoogle.com/periodic-table/) to discuss the concept of valence electrons and the octet rule.

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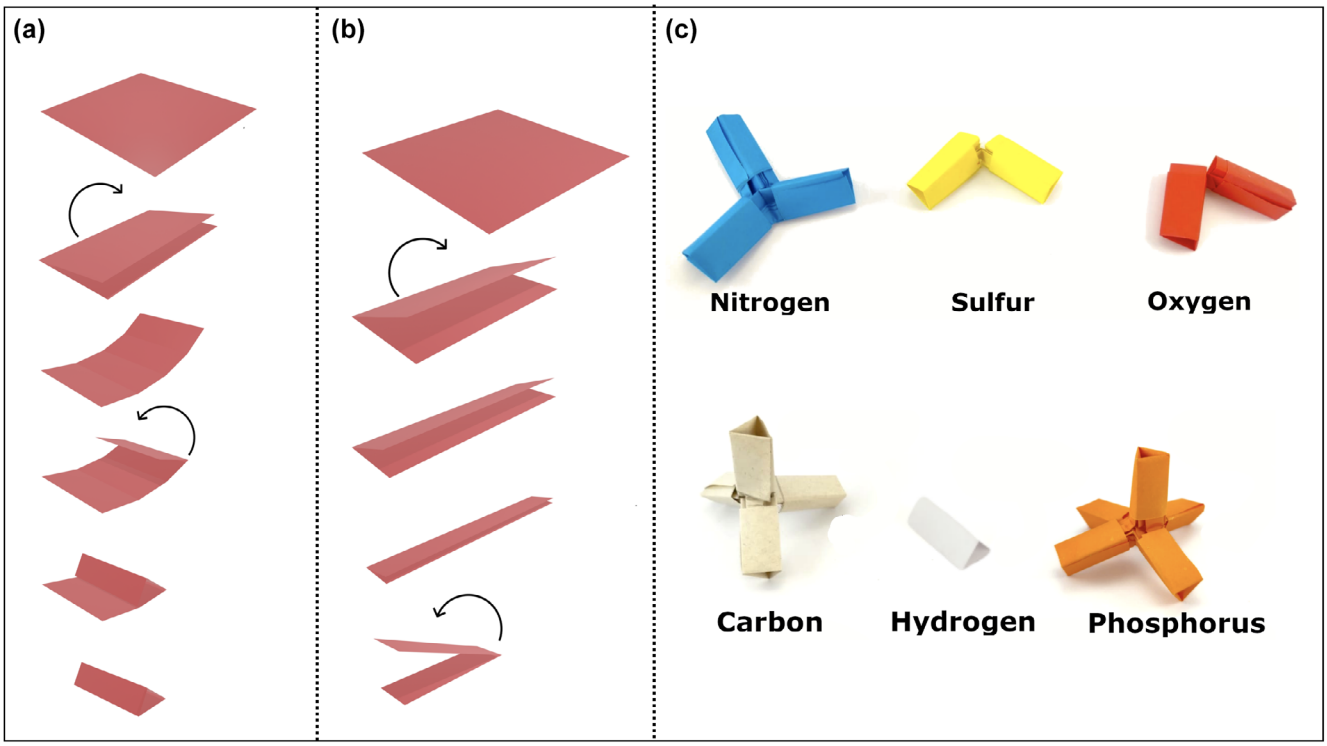


FIGURE 1 Folding instructions for building blocks. (a) Prism, (b) connector, and (c) the six primary bioelements built using our proposal. Nitrogen, sulfur, oxygen, carbon, hydrogen, and phosphorus. Each prism represents the number of electrons available for chemical bonding.

With this information the game is repeated for the construction of each of the six primary bioelements, using the prisms and connectors. Thus, hydrogen is made with a single white prism, oxygen contains two red prisms bound by one connector, nitrogen has three blue prisms and two connectors, carbon has four gray prism and three connectors, phosphorus is made with five orange prisms and four connectors and sulfur contains two yellow prisms and one connector (Figure 1c). At the end of the class the student that gets the highest score obtains the maximum grade on the activity. Column fourth on Table 1 shows the maximum number of points available for the activity.

1.2.2 | Second session: Building simple biomolecules from previously built bioelements

Prior to this activity students are requested to bring to class building blocks of each color 50 prisms, 80 connectors and square paper (5 × 5 cm) of each color. The game consists of a group competition starting by building a carbon atom following the same game rules used in session 1. Next, students are asked to bind hydrogen prisms to the newly built carbon atom. Students realize they do not have the tools to link both atoms. This creates a learning moment in which students reflect on the function of chemical bonds and their physical nature to generate a molecule. Students are introduced to instructions displayed on a screen using the image on Figure 2a. The instructor explains that this is called a link, and represents the chemical bond between two atoms. As an example, Figure 2b–g shows how to build a simple molecule of water, Figure 2b,c shows how electrons are bound using a connector. Figure 2d shows how the atom of oxygen is represented, Figure 2e,f shows the links inserted into the prisms, and finally, Figure 2g shows the hydrogen prisms bound to the oxygen atom. Students are then asked to build the six simple molecules described in column 2, Table 1.

Once the students finished, it is requested to them to compare the origami-built structure to the structure generated by the tool denominated "molecular panel" developed in the Universidad Autónoma de Madrid (http://qorganica.es/QOT/T00/inicio_exported/index.html#).

1.2.3 | Third session: Building complex biomolecules from previously built simple molecules

Prior to this activity students are asked to bring to class the simple molecules built on session 2 and additional building blocks of different colors. The game consists of a team competition to build the simplest amino acid

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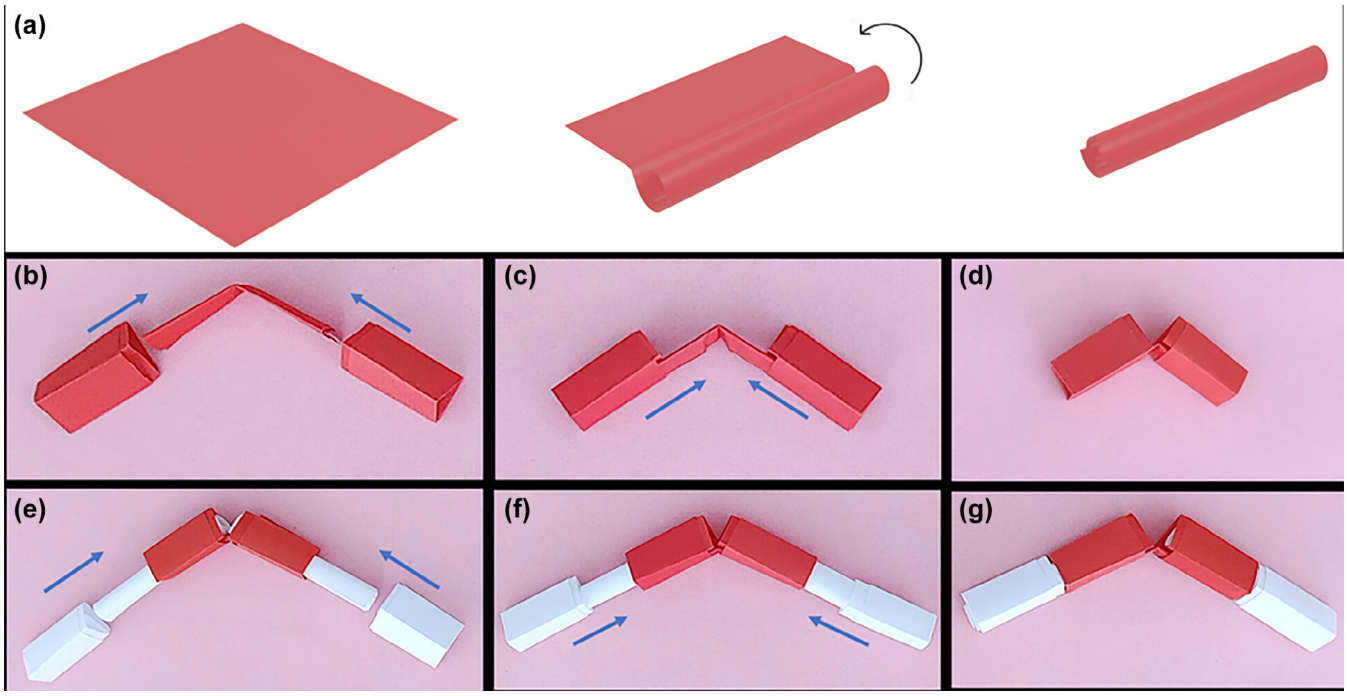


FIGURE 2 Folding instructions for building blocks (a). Link (b–g) Making a molecule of water with origami building blocks. (b,c) Two red prisms (electrons) are bound using a connector (arrows). (d) The atom of oxygen is thus represented with two prisms bound with a connector. (e,f) Links (arrows) are inserted into the prisms. (g) Two hydrogen prisms (white) are linked to the oxygen atom to create water (H_2O).

(glycine) shown on a screen using Figure 3a (2D molecular structure). The team that finished first with the correct structure (Figure 3b–d) gets one point. The instructor uses the model to demonstrate the position of the α-carbon and the description of the amino and carboxylic acid functional groups. Then, each team is assigned with one of the five molecules presented in column 2 of Table 1. Students are given instructions on how to search for and to visualize 2D and 3D structures of their assigned molecule using the open access chemical database "Chemspider" (Table 1). At the end of the class the first group to complete the assignment and correctly explain basic questions about the molecule's chemical structure earns one point and receives the highest grade. Students are also asked to compare their origami models to the 3D structures in Chemspider. This is another interesting learning moment in which students realize how much their origami molecular structures resemble the geometric configuration of the database's 3D-structural models (Figure S1).

1.3 | Feedback from students

We anonymously surveyed 12 groups of 3–4 students to evaluated the usability and perceived utility of the strategy using a single question "Do you think this strategy was useful to understand 3D structure of bioelements and their importance on building simple and complex biomolecules?" Each answer was analyzed through content analysis that included: (i) identifying the general idea conveyed in the response (ii) listing key adjectives and verbs that described the main idea in a matrix (Table S1). (iii) submitting the matrix with the adjectives and verbs through a frequency analysis on the Word Cloud generator extension of Google to obtain the word cloud shown in Figure 4.

Overall, students qualified origami as a favorable strategy to learn molecular structure, none of the teams expressed a negative perception towards the utility or use of this activity in their learning. They highlighted that it was easy and useful. Students referred that the instructions for the building blocks were simple, building a block was faster than expected, and aesthetically they perceived the process and the final product as beautiful. In addition, students highlighted that origami was useful to grasp concepts such as electromagnetic forces, valence electrons or chemical bonds. Students valued competition and gamification elements as challenging and fun that motivated them to continue through the learning process.

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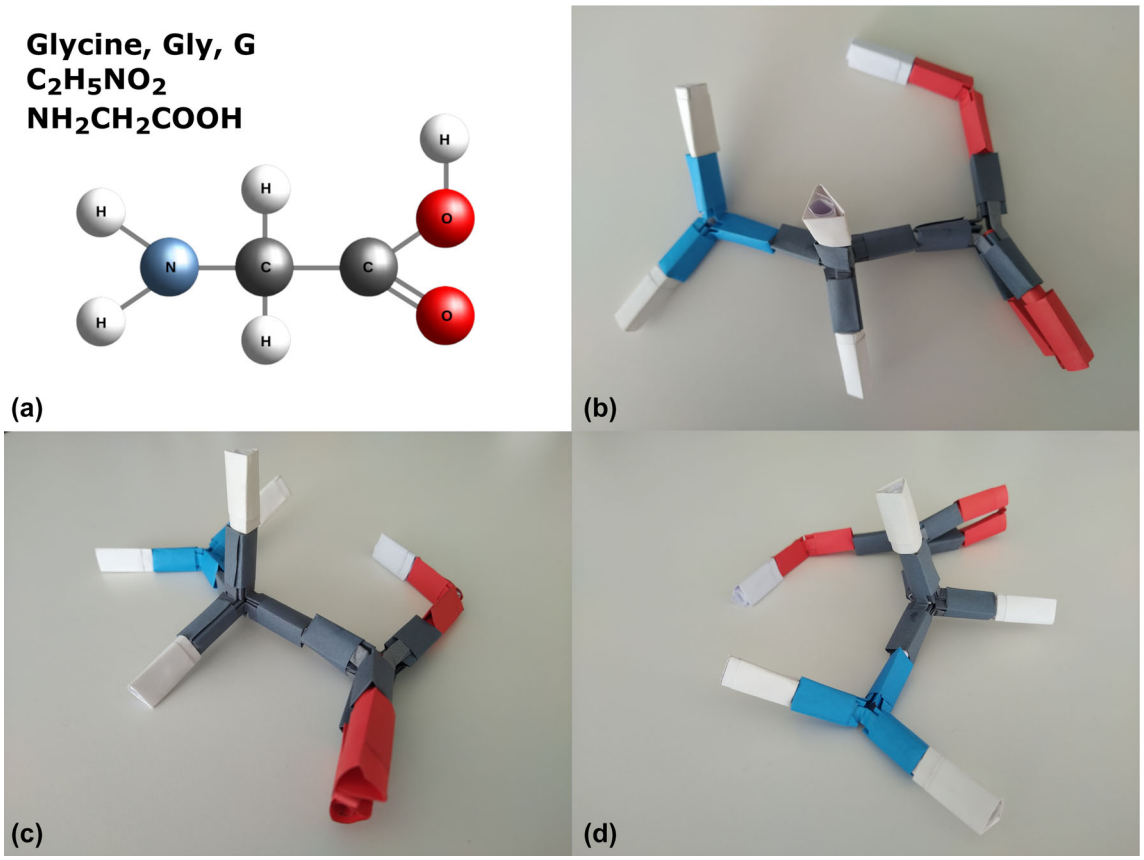


FIGURE 3 Examples of the visual representations of the structure of a biomolecule, the amino acid Glycine. (a). Different representations of glycine starting with the name (Glycine), three-letter code (Gly), single letter code (G); molecular formula $C_2H_5NO_2$ (describes the number of atoms for the biomolecule), condensed chemical formula NH_2CH_2COOH (the distribution of bioelements in the biomolecule), and structural formula in 2D. (b–d) 3D structure of the amino acid using our proposal. It allows us to visualize the angular position of bioelements otherwise not observed in the 2D formula. (b). 3D structure observed as a projection of the 2D structure in A. (c). 3D-view from the carboxyl-group. (d) 3D-view from the amino-group.

1.4 | Final considerations from implementing origami to teach molecular structure

In our experience, leading the topic of molecular structure with origami was amenable and challenging, and generated expectations about students' reception and performance. Moreover, we believe that the strategy allowed students to feel that they achieved the proposed learning objectives, as it was clearly expressed by students' team assessment. After reflecting about the implementation of this experience, we would recommend the following aspects that we did not consider and that might improve the experience to obtain better results. Time may be a limiting factor depending on students' skills. It is recommended to perform a prior activity to identify students with different levels of visual literacy and solid knowledge on general chemistry. Those with high skills could be the leaders from which teams are formed. This would motivate the leader and the whole team to support the representative and make more competitive teams, generating a motivating environment in which everyone has similar chances to win. Also, the preliminary results presented here, motivate us to consider including evaluation activities/questions within the course to verify that students are indeed learning about molecular structure when they use origami. This would allow us to prepare a consistent assessment of how origami facilitates to achieve the learning objectives.

Finally, although molecular origami's proposal by Garrido already developed a conceptual framework for what different building blocks represents (tetrahedral atoms regardless of their valence electrons), we believe that representing atoms by their most common valence electrons improved the original model allowing us to integrate other concepts such as the electromagnetic and intermolecular forces. Furthermore, since our origami models resemble the geometry of 3D structures of

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FIGURE 4 Word Cloud based on the students' perception survey assessing usability and utility of origami to learn molecular structure of biomolecules. Students described the activity using several adjectives and verbs. The verb "grasp" was the most frequent word to describe origami as a useful strategy to learn molecular structure.

molecules, we suggest that origami might be used to explore additional concepts such as molecular geometry. So far, we did not explore this pathway, but readers may be aware of different strategies to use origami for this purpose.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

All data analyzed is provided within this article. Additional information such as the response to the perception survey might be shared after contacting the corresponding author.

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